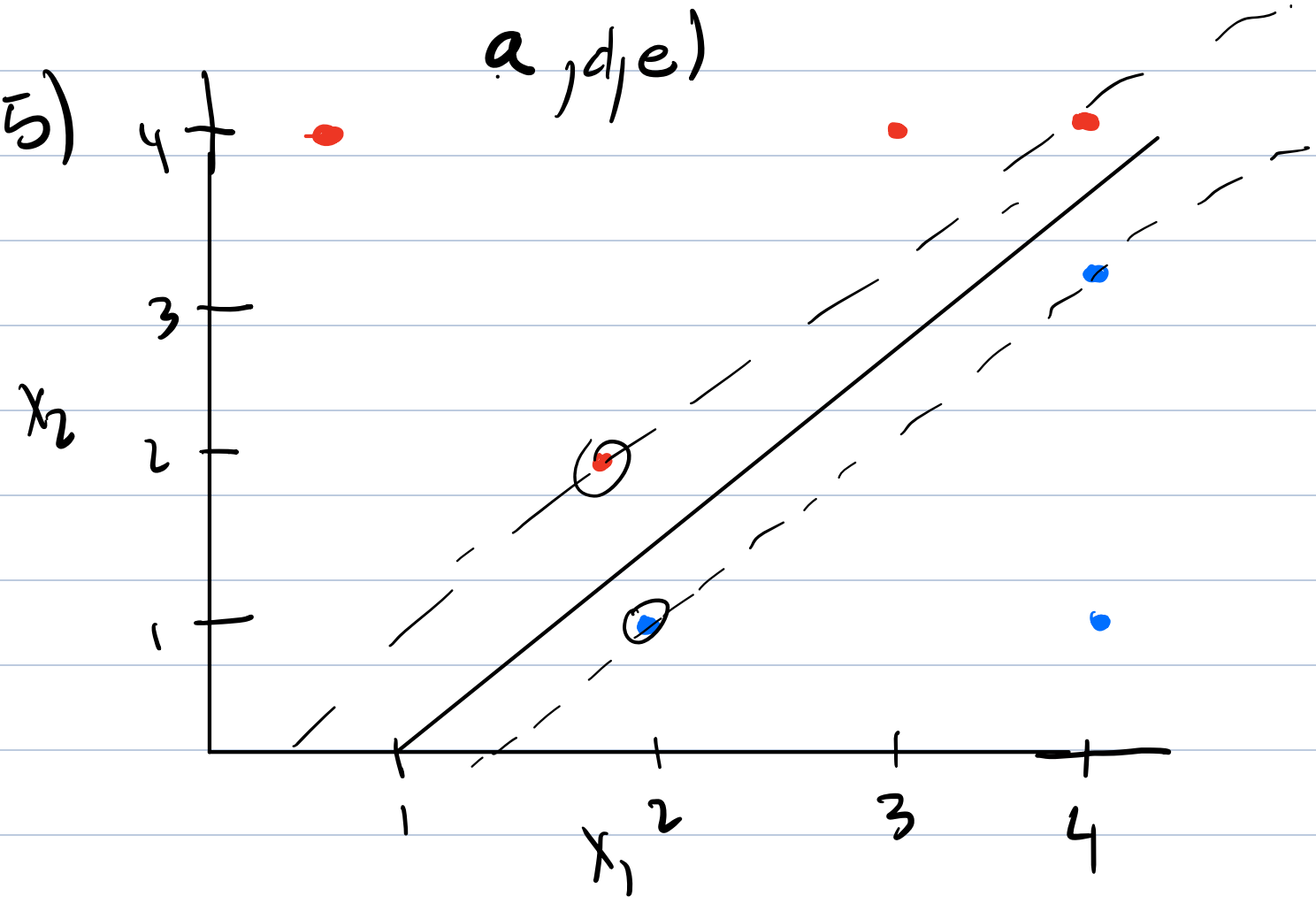
HW 6

5) a, d, e)



b) Assume that the hyperplane passes through the midpoint of (2,1) and (2,2) → (2, 1.5)

you can get an equation like

$y = x - 0.5$ or $x_2 = x_1 - 0.5$

c) $0 = X_1 - X_2 - 0.5$

$-1.5 = 3 - 4 - 0.5$ Obs 1 → Red

$-0.5 = 2 - 2 - 0.5$ Obs 2 → Red

$0.5 = 2 - 1 - 0.5$ Obs 5 → Blue

$0.5 = 4 - 3 - 0.5$ Obs 6 → Blue

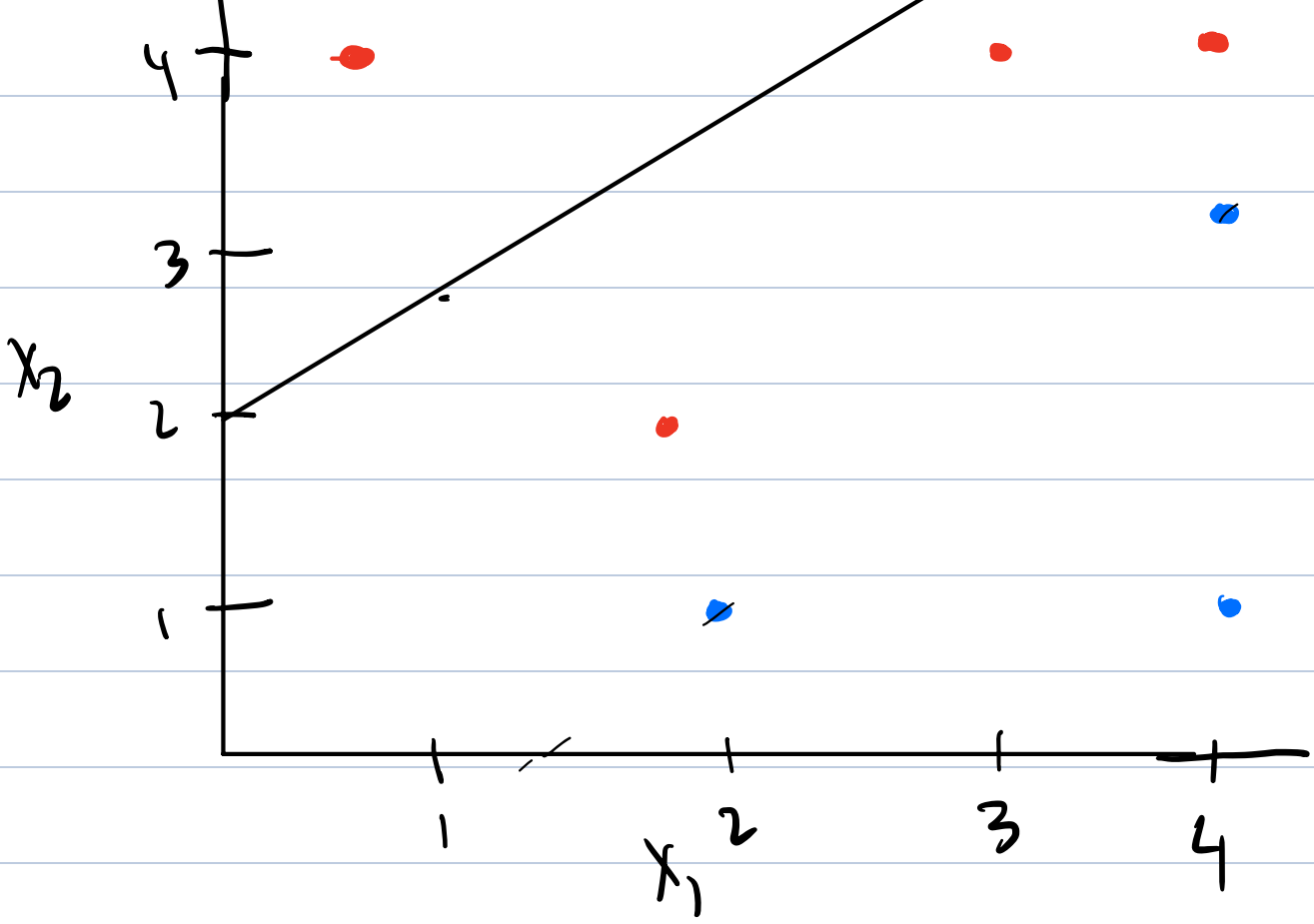
Classify to Red if $X_1 - X_2 - 0.5 < 0$

Classify to Blue if $X_1 - X_2 - 0.5 > 0$

$\beta_0 = -0.5$ $\beta_1 = 1$ $\beta_2 = -1$

f) the seventh observation is clearly outside the margin and it is not a support vector so observation 7 changing slightly would not change anything

g)



$y = 2 + x$

$x_2 = 2 + x_1$

$0 = 2 + x_1 - x_2$

h)

